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(54) BATTERY MANUFACTURING APPARATUS AND BATTERY MANUFACTURING METHOD OF THE SAME

(71) Applicant: SK On Co., Ltd., Seoul (KR)

(72) Inventors: Yong Hun JUNG, Daejeon (KR); Wook Hyun KIM, Daejeon (KR)

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CPC H01M 50/516 (2021.01); H01M 50/505 (2021.01)

(57) ABSTRACT

A battery manufacturing apparatus for manufacturing a battery assembly including a plurality of battery cells, each having a lead tab portion electrically connected to an outside and protruding outwardly, and a busbar electrically connecting one or more battery cells among the plurality of battery cells of the present disclosure includes a welding wire, at least a portion of which is melted to weld the lead tab portion and the busbar, a supply portion supplying the welding wire to the lead tab portion, a guiding portion contacting the welding wire discharged from the supply portion and moving the welding wire in a direction in which the lead tab portion extends, and a laser irradiation portion irradiating the lead tab portion, the busbar, or the welding wire with a laser.

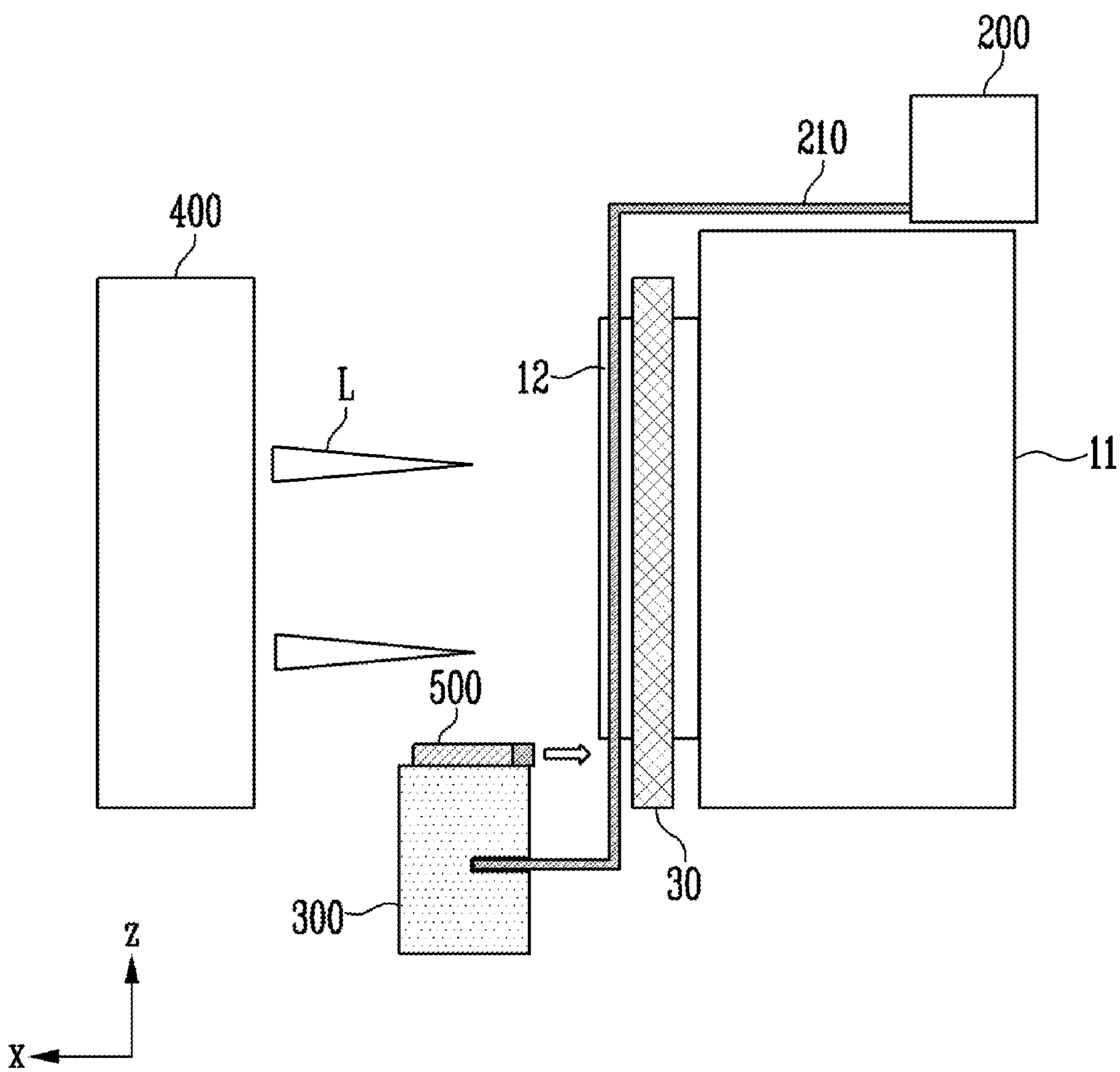


FIG. 1

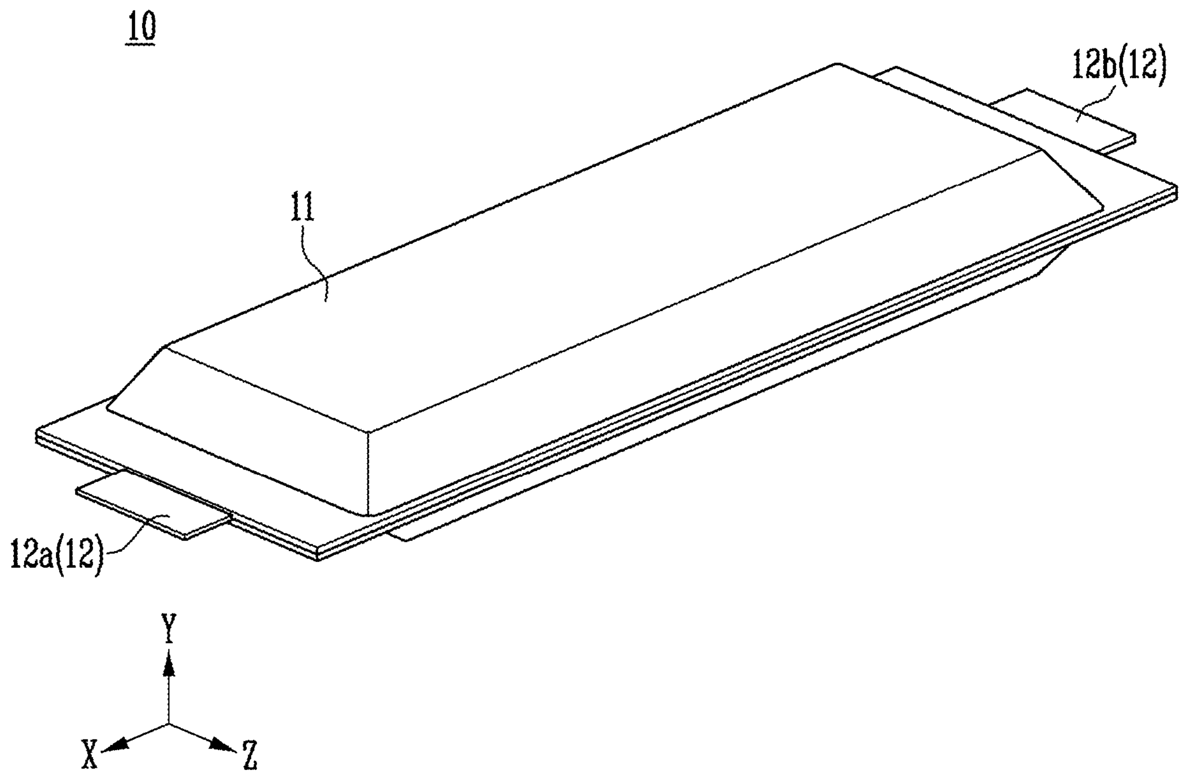


FIG. 2

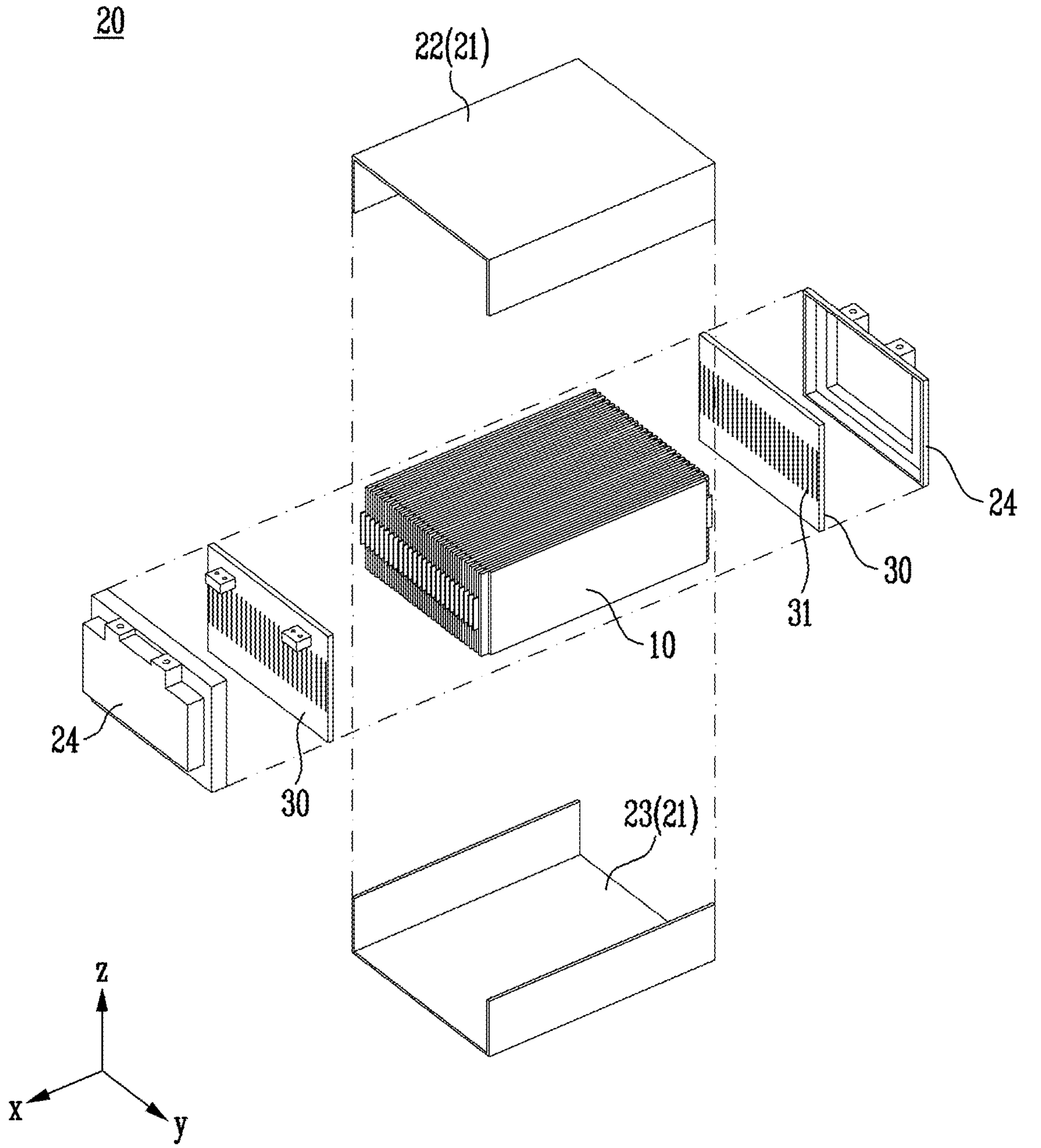


FIG. 3

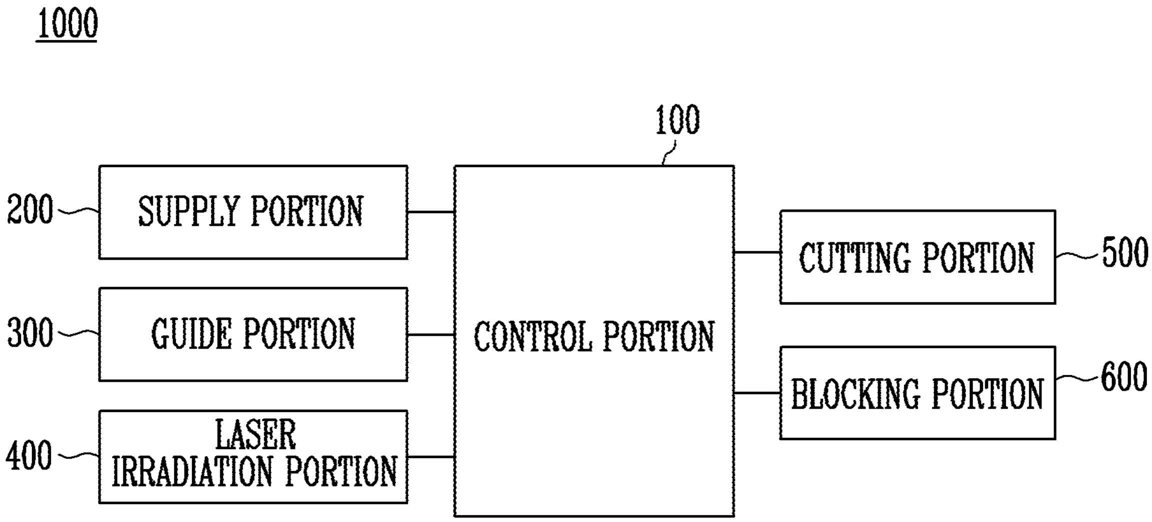

1000
100
200
SUPPLY PORTION
300
GUIDE PORTION
400
LASER
IRRADIATION PORTION
CONTROL PORTION
CUTTING PORTION
500
BLOCKING PORTION
600

FIG. 4

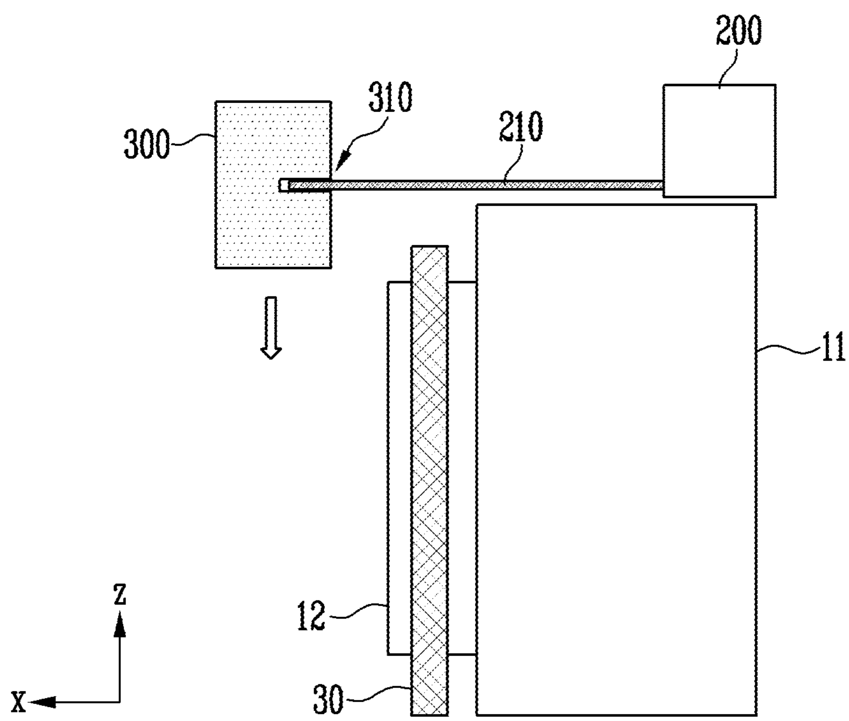

200
310
210
300
11
12
30
Z
X

FIG. 5

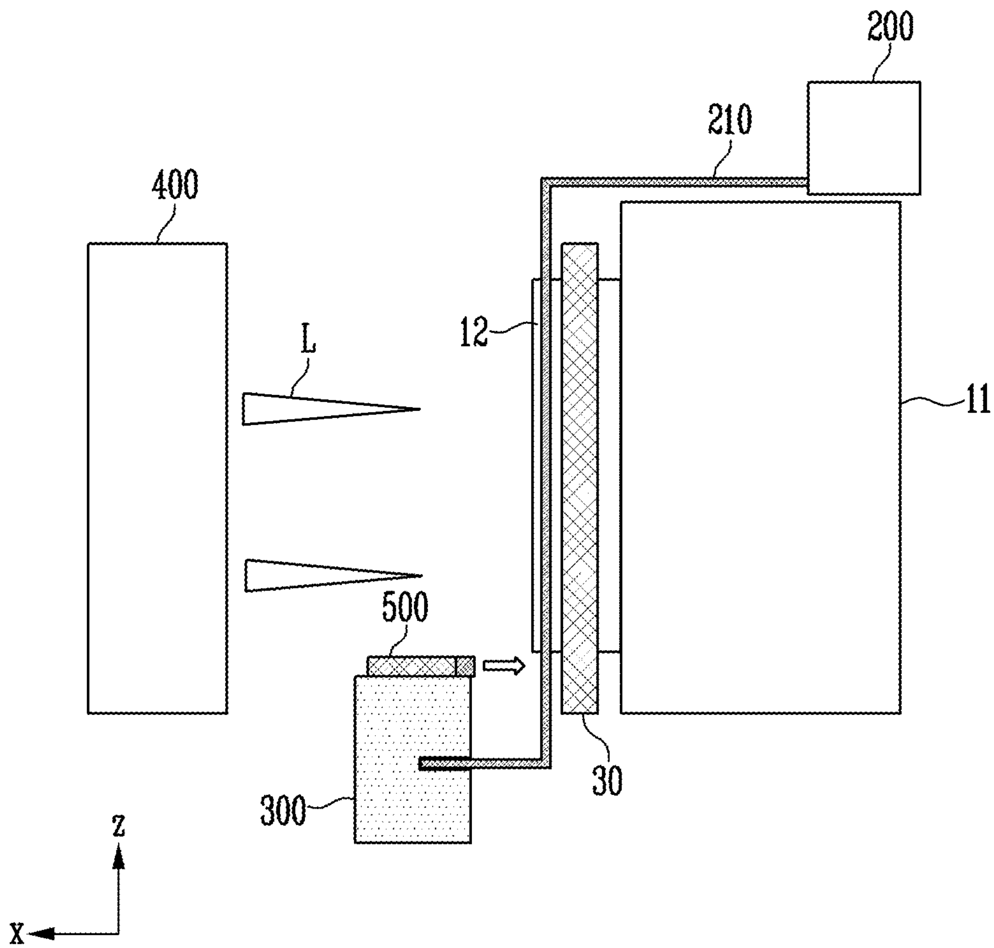


FIG. 6

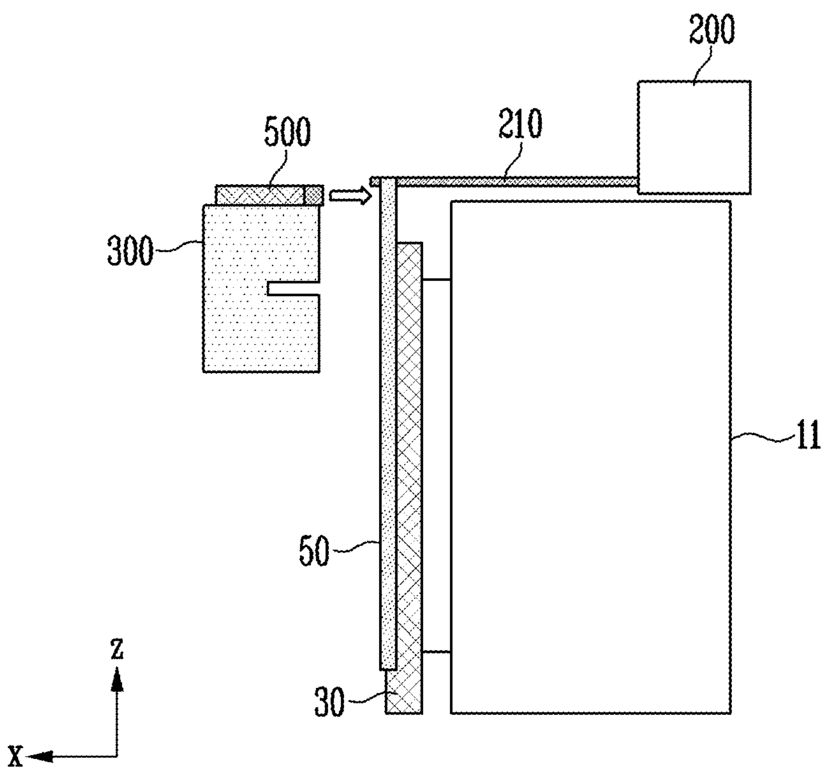


FIG. 7

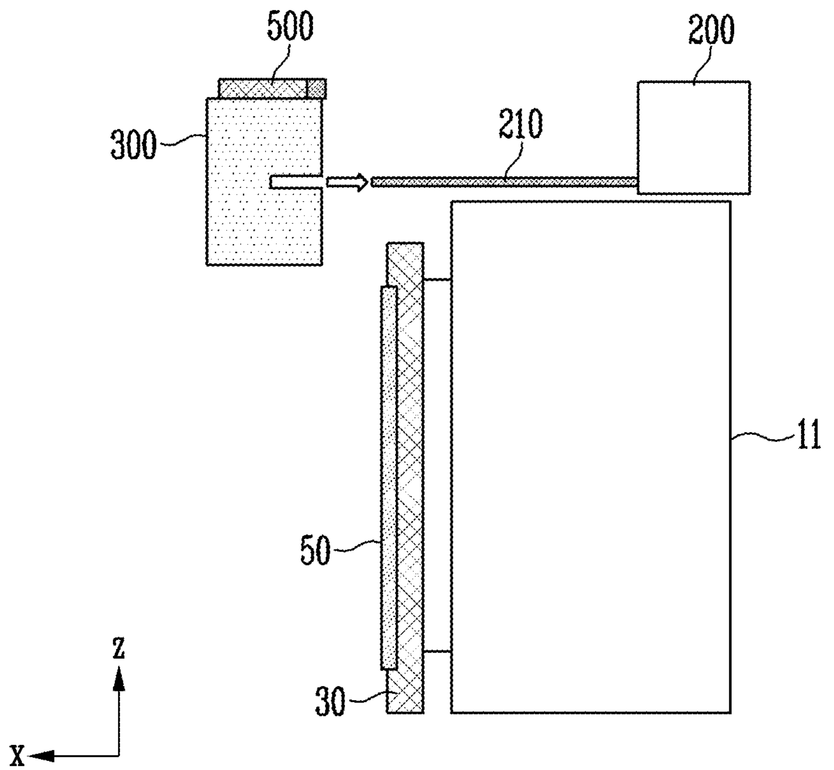


FIG. 8

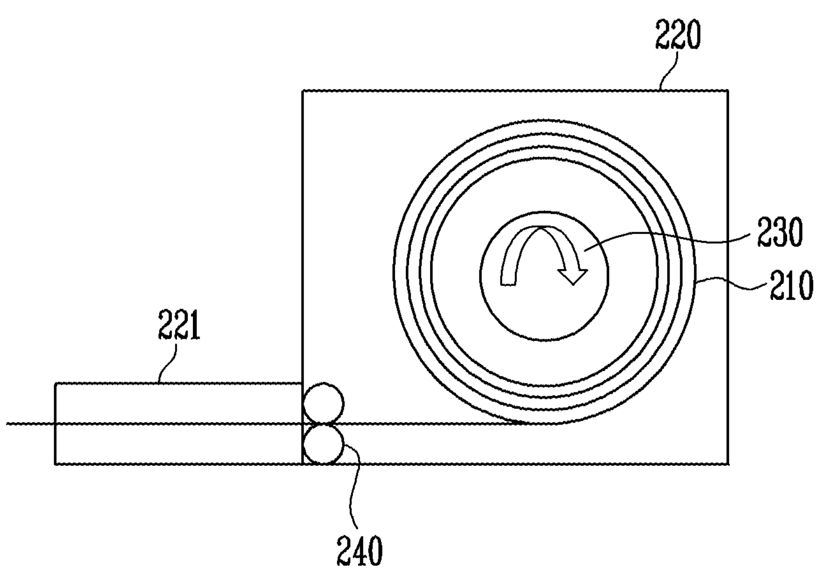


FIG. 9

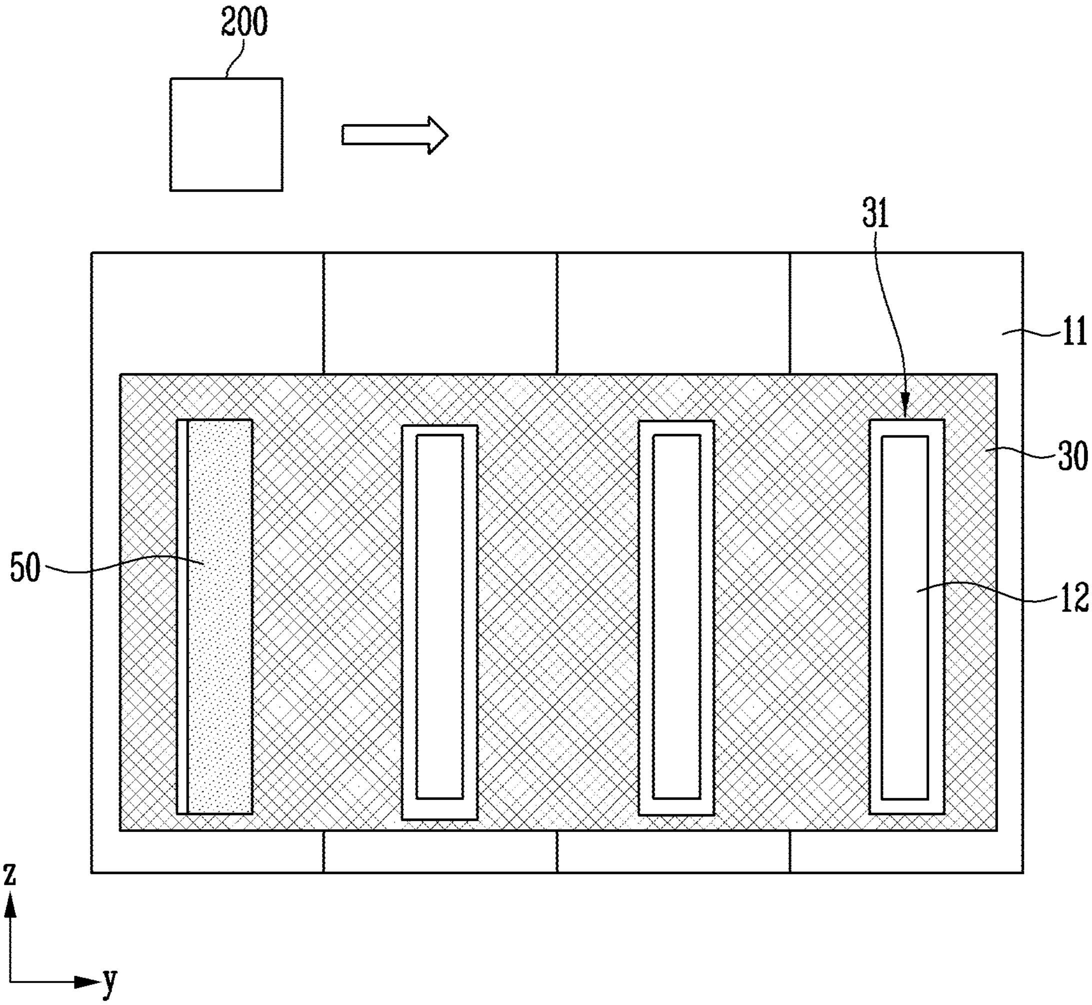


FIG. 10

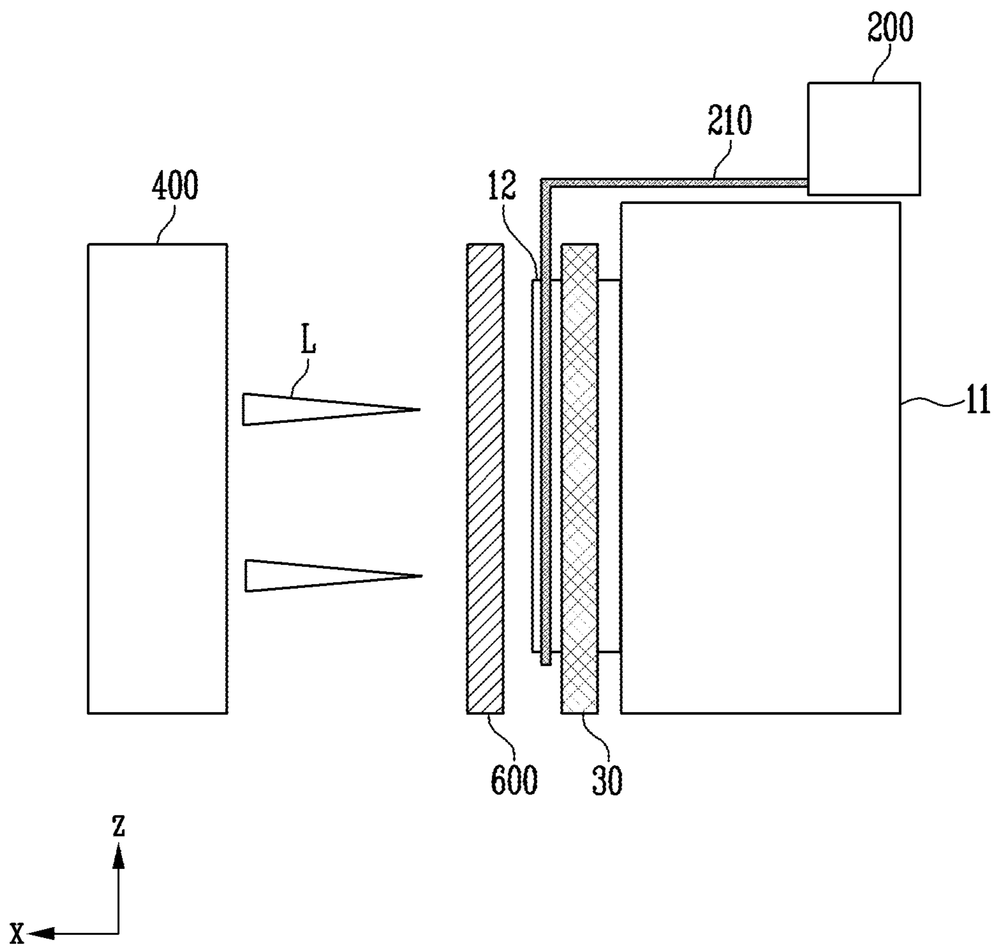


FIG. 11

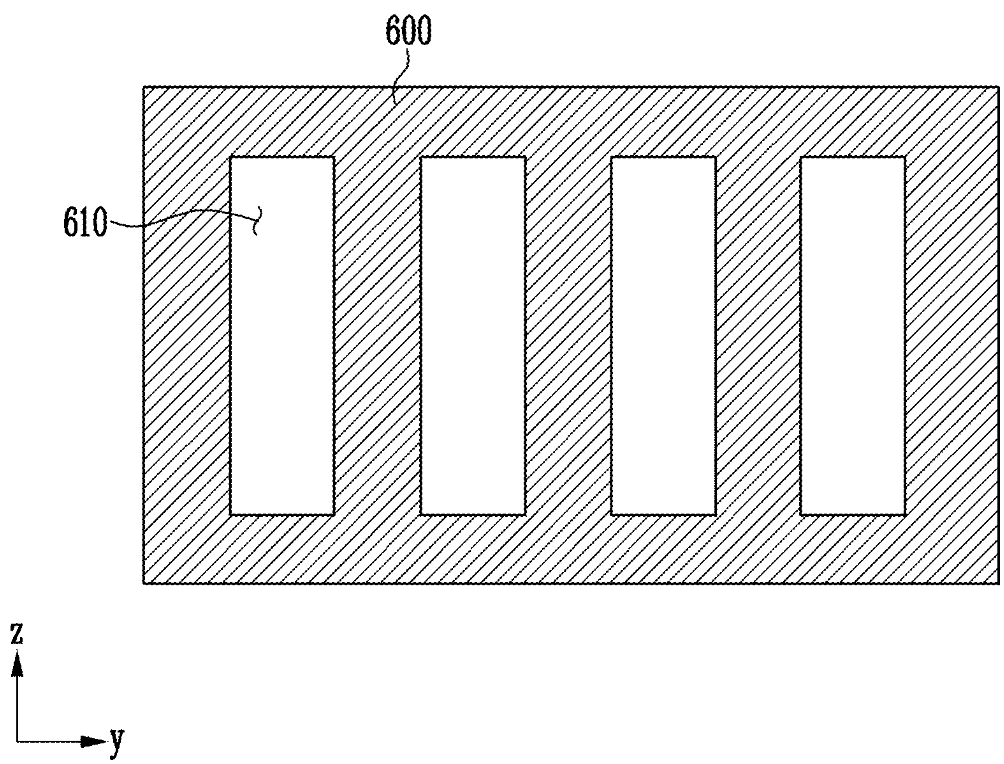
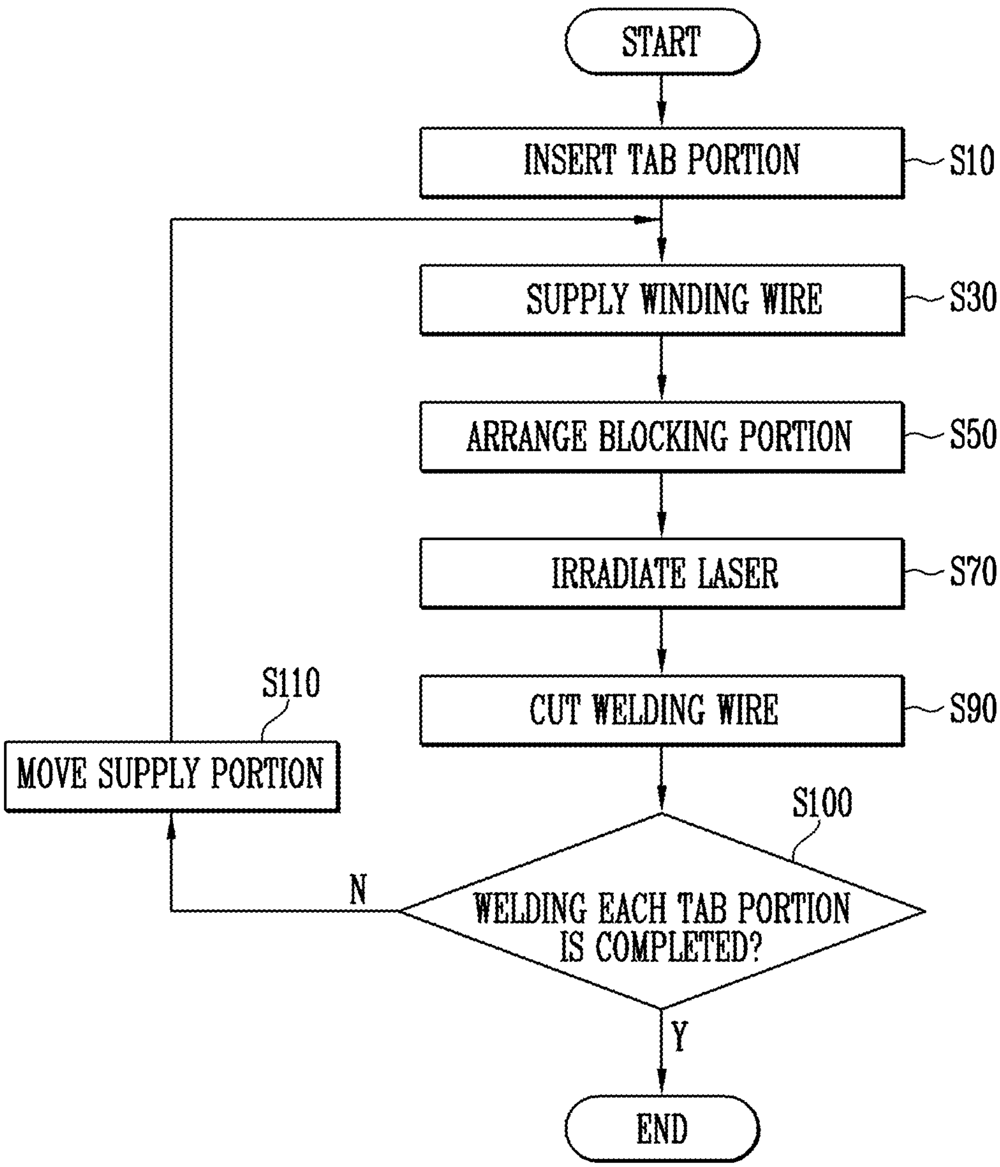


FIG. 12


START
INSERT TAB PORTION
S10
SUPPLY WINDING WIRE
S30
ARRANGE BLOCKING PORTION
S50
IRRADIATE LASER
S70
CUT WELDING WIRE
S90
S110
MOVE SUPPLY PORTION
S100
N
WELDING EACH TAB PORTION IS COMPLETED?
Y
END

BATTERY MANUFACTURING APPARATUS AND BATTERY MANUFACTURING METHOD OF THE SAME

CROSS-REFERENCE TO RELATED PATENT APPLICATION

[0001] The present application claims priority under 35 U.S.C. § 119 (a) to Korean patent application number 10-2024-0030762 filed on Mar. 4, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

1. Field

[0002] The present disclosure relates to a battery manufacturing apparatus and a battery manufacturing method using the battery manufacturing apparatus.

2. Description of the Related Art

[0003] A secondary battery converts electrical energy into chemical energy and stores the chemical energy, and can be reused multiple times by charging and discharging. Secondary batteries are widely used in various industries due to their economical and eco-friendly characteristics. In particular, lithium secondary batteries are widely utilized across industries, including mobile devices which require high-density energy.

[0004] The operating principle of a lithium secondary battery is an electrochemical oxidation-reduction reaction. In other words, electricity is generated by the movement of lithium ions and charged through the opposite process. In the case of a lithium secondary battery, the phenomenon of lithium ions escaping from an anode and moving to a cathode through an electrolyte and a separator is referred to as discharging. The opposite process of the phenomenon is referred to as charging.

[0005] To increase the capacity and output of secondary batteries, the secondary batteries may be bundled together. A busbar may be used to electrically connect the respective secondary batteries. When the secondary battery is welded to the busbar, the welding performance varies depending on a status of the secondary battery or the busbar, which affects the performance of the secondary battery. Therefore, research is being actively conducted to improve the welding performance of the secondary battery and the busbar.

SUMMARY OF THE INVENTION

[0006] An object of the present disclosure is to weld a battery cell and a busbar quickly and reliably.

[0007] Another object of the present disclosure is to performing welding reliably regardless of a length of a lead tab portion.

[0008] In addition, the present disclosure can be widely applied in the field of electric vehicles, battery charging stations, and green technology, such solar power generation, and wind power generation using batteries.

[0009] The present disclosure can be used in eco-friendly electric vehicles, hybrid vehicles, etc. to prevent climate change by suppressing air pollution and greenhouse gas emissions.

[0010] A battery manufacturing apparatus for manufacturing a battery assembly including a plurality of battery cells, each having a lead tab portion electrically connected to an outside and protruding outwardly, and a busbar electrically connecting one or more battery cells among the plurality of battery cells according to embodiments of the present disclosure includes a welding wire, at least a portion of which is melted to weld the lead tab portion and the busbar, a supply portion supplying the welding wire to the lead tab portion, a guiding portion contacting the welding wire discharged from the supply portion and moving the welding wire in a direction in which the lead tab portion extends, and a laser irradiation portion irradiating the lead tab portion, the busbar, or the welding wire with a laser.

[0011] The supply portion may include a winder around which the welding wire is wound on an outer side thereof, and a roller unidirectionally transferring the welding wire unwound from the winder by rotation with the welding wire interposed therebetween.

[0012] The winder may be rotatably provided.

[0013] The guiding portion further may include a fixing portion formed by recessing one surface thereof.

[0014] The fixing portion may be removable from the welding wire.

[0015] The fixing portion may be movable in the direction in which the lead tab portion extends.

[0016] The battery manufacturing apparatus may further include a cutting portion cutting the welding wire moved in the direction in which the lead tab portion extends.

[0017] The cutting portion may be located on the guiding portion.

[0018] The cutting portion may be a knife.

[0019] The battery manufacturing apparatus may further include a blocking portion positioned between the laser irradiation portion and the plurality of battery cells and covering one or more battery cells among the plurality of battery cells.

[0020] The blocking portion may be formed by penetrating a region corresponding to the lead tab portion.

[0021] The supply portion may be movable in a direction in which the plurality of battery cells are stacked.

[0022] A battery manufacturing method of manufacturing a battery assembly including a plurality of battery cells, each having a lead tab portion electrically connected to an outside and protruding from one side thereof, and a busbar electrically connecting one or more battery cells among the plurality of battery cells battery manufacturing method includes inserting the lead tab portion into the busbar, supplying the welding wire toward the lead tab portion from the supply portion supplying the welding wire welding the lead tap portion and the busbar by melting at least a portion of the welding wire, and irradiating the lead tab portion, the busbar, or the welding wire with a laser.

[0023] In the supplying of the welding wire, the welding wire may move in a direction in which the lead tab portion extends.

[0024] The battery manufacturing method may further include moving the supply portion in a direction in which the plurality of battery cells are stacked.

[0025] The battery manufacturing method may further include arranging a blocking portion covering one or more battery cells among the plurality of battery cells on one side of the plurality of battery cells in a protruding direction of the lead tab portion.

[0026] The irradiating the laser may include irradiating the laser in a direction in which the lead tab portion extends.

[0027] The battery manufacturing method may further include cutting the welding wire moved in a direction in which the lead tab portion extends.

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. **1** shows a battery cell according to one embodiment of the present disclosure.

[0029] FIG. **2** shows a battery assembly according to one embodiment of the present disclosure.

[0030] FIG. **3** is an example block diagram related to control of a battery manufacturing apparatus according to one embodiment of the present disclosure.

[0031] FIGS. **4** to **7** show a welding process of a battery manufacturing apparatus according to one embodiment of the present disclosure.

[0032] FIGS. **8** and **9** show a supply portion according to one embodiment of the present disclosure.

[0033] FIGS. **10** and **11** show a blocking portion according to one embodiment of the present disclosure.

[0034] FIG. **12** is a flowchart illustrating a battery manufacturing method according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

[0035] Hereinafter, the present disclosure will be described in detail with reference to the attached drawings. This is merely illustrative, and the present disclosure is not limited to the specific embodiments described in an illustrative manner.

[0036] FIG. **1** shows a battery cell **10** according to one embodiment of the present disclosure, and FIG. **2** shows a battery assembly **20** according to one embodiment of the present disclosure.

[0037] The battery cell **10** described herein refers to a secondary battery which can be used repeatedly by charging and discharging electrical energy. In one example, the battery cell **10** may refer to a lithium secondary battery or a lithium-ion battery. However, the present disclosure is not limited thereto. In another example, the battery cell **10** may refer to a solid-state battery.

[0038] The battery cell **10** may be classified into as a pouch secondary battery, a prismatic secondary battery, or a cylindrical secondary battery based on the shape thereof. For ease of explanation, a prismatic secondary cell is shown herein as an example, but the present disclosure is not limited thereto.

[0039] The battery assembly described herein may refer to a battery module or a battery pack. The battery module may refer to a group of one or more battery cells **10** received in a case to be protected from external shock, heat, vibration, etc. for high power and high capacity characteristics. The battery pack may refer to a group of one or more battery cells or battery modules.

[0040] Referring to FIG. **1**, the battery cell **10** includes a lead tab portion **12** which is electrically connected to the outside and protrudes outwardly. The battery cell **10** may further include a body portion **11**.

[0041] The body portion **11** may produce or store electrical energy. The body portion **11** may include an electrode assembly (not shown) which includes an anode, a cathode, and a separator for producing and storing electrical energy. The body portion **11** may further include an electrolyte in contact with the electrode assembly. The electrolyte may be an electrolyte liquid provided as a liquid.

[0042] The body portion **11** may further include an exterior material. The exterior material may have an electrode assembly and an electrolyte liquid therein. The exterior material may be a highly rigid material to protect the electrode assembly and the electrolyte liquid received therein from impact.

[0043] The lead tab portion **12** may electrically connect the body portion **11** to the outside. The lead tab portion **12** may include a plurality of lead tab portions. The lead tab portion **12** may include an anode tab **12***a* connected to an anode and a cathode tab **12***b* connected to a cathode. More specifically, the lead tab portion **12** may include the anode tab **12***a* connected to at least one anode plate, and the cathode tab **12***b* connected to at least one cathode plate.

[0044] The lead tab portion **12** may protrude in a direction away from both sides of the body portion **11**. Referring to FIG. **1**, the anode and cathode tabs may protrude in opposite directions. However, the present disclosure is not limited thereto. The anode tab and the cathode tab may be provided next to each other on one surface of the body portion **11**.

[0045] As used herein, the protruding direction of the lead tab portion **12** may refer to a direction parallel to the X-axis direction. Further, the direction in which the lead tab portion **12** extends may refer to a direction parallel to the Z-axis direction. Further, the lead tab portion **12** may refer to an electrode lead tab.

[0046] More specifically, FIG. **2** is an exploded view of the battery assembly **20**. The battery assembly **20** may include a plurality of battery cells **10**. The plurality of battery cells **10** may be stacked in a first direction. For example, the plurality of battery cells **10** may be stacked in a y-axis direction.

[0047] An adhesive member (not shown) may be provided between the plurality of battery cells **10**. The plurality of battery cells **10** may be reliably connected by the adhesive member. For example, the adhesive member may be a tape. Further, a battery compression pad (not shown) may be provided between the plurality of battery cells **10**. The compression pad may relieve the surface pressure between the battery cells **10** to prevent damage to the battery cells **10**. To this end, the compression pad may include a shock absorbing material or an elastic material.

[0048] In this specification, the direction in which the plurality of battery cells **10** are stacked may refer to a direction next to the Y-axis direction.

[0049] The battery assembly **20** may include a receiving case **21** forming a receiving space in which the plurality of battery cells **10** are housed. In other words, the plurality of battery cells **10** may be located in the receiving space.

[0050] The receiving case **21** may include a lower cover **23** which supports the plurality of battery cells **10** and an upper cover **22** which is engaged with the lower cover **23** and covers the receiving space. Referring to FIG. **2**, the lower cover **23** and the upper cover **22** may be coupled together to accommodate the plurality of battery cells **10** therein.

[0051] More specifically, the plurality of battery cells **10** may be inserted through an opening in the lower cover **23**. After the plurality of battery cells **10** are positioned on the lower cover **23**, the upper cover **22** may be coupled with the lower cover **23** to cover the plurality of battery cells **10**.

[0052] The battery assembly **20** may further include an end cover **24**. The end cover **24** may be coupled to the receiving case **21** to cover the receiving space. The end cover **24** may protect the battery cells **10** located in the receiving space. To this end, the end cover **24** may be coupled to the receiving case **21**. For example, the end cover **24** and the receiving case **21** may be connected by welding or bolting.

[0053] The battery assembly **20** may further include a busbar **30**. The busbar **30** may electrically connect at least some of the plurality of battery cells **10**. The busbar **30** may be positioned to face the battery cells **10** in the protruding direction of the lead tab portions **12**. Referring to FIG. **2**, the busbar **30** may be positioned on either side of the battery cells **10**.

[0054] The lead tab portion **12** and the busbar **30** may be electrically connected to each other. For simplicity and reliability, the lead tab portion **12** may be inserted into an insertion hole **31** provided in the busbar **30**. The insertion hole **31** may be formed through one surface of the busbar **30**. As the busbar **30** approaches the plurality of battery cells **10**, one end of the lead tab portion **12** may be inserted into the insertion hole **31**.

[0055] The busbar **30** may include an insertion slit which is provided in the form of a slit rather than a hole. The lead tab portion **12** may be inserted through an opening in the insertion slit. The present disclosure is not limited thereto. The busbar **30** may have a variety of shapes into which the lead tab portion **12** is inserted.

[0056] When the lead tab portion **12** is inserted into the busbar **30**, a portion of the lead tab portion **12** may protrude outwardly. When the lead tab portion **12** and he busbar **30** are welded, the weld performance may vary depending on the protruding length of the lead tab portion **12**. For example, when the lead tab portion **12** is short, a weldable area with the busbar **30** is reduced, which may result in poorer weld quality.

[0057] The disclosure may include the welding wire **210** which is at least partially molten to weld the lead tab portion **12** and the busbar **30**. More specifically, the present disclosure may provide the welding wire **210** which allows a welded joint to be reliably formed between the lead tab portion **12** and the busbar **30**.

[0058] FIG. **3** is an example block diagram related to control of a battery manufacturing apparatus according to one embodiment of the present disclosure.

[0059] Referring to FIG. **3**, the battery manufacturing apparatus of the present disclosure includes a supply portion **200**, a guiding portion **300**, and a laser irradiation portion **400**. The supply portion **200** supplies the welding wire **210** which is at least partially molten to weld the lead tab portion **12** and the busbar **30**.

[0060] The guiding portion **300** may contact the welding wire **210** which is discharged from the supply portion **200** to move the welding wire **210** in the direction in which the lead tab portion **12** extends. The laser irradiation portion **400** irradiates the lead tab portion **12**, the busbar **30**, or the welding wire **210** with a laser.

[0061] The battery manufacturing apparatus of the present disclosure further includes a control portion **100**. The control portion **100** may control the supply portion **200**, the guiding portion **300**, and the laser irradiation portion **400**. The present disclosure may further include a cutting portion **500** or a blocking portion **600** to be described below. The control portion **100** may control the cutting portion **500** or the blocking portion **600**. In the following, each configuration will be described in detail with reference to the drawings.

[0062] FIGS. **4** to **7** schematically show a welding process of a battery manufacturing apparatus according to one embodiment of the present disclosure.

[0063] Referring to FIG. **4**, the supply portion **200** may supply the welding wire **210** to the lead tab portion **12**. The welding wire **210** may include a filler metal. The welding wire **210** may be used to regulate heat capacity during welding. The welding wire **210** may be used to weld two metals which do not react with each other, or may be added to materials which react with each other for welding.

[0064] At least a portion of the welding wire **210** may be melted. The laser irradiated during welding may melt a portion of the welding wire **210**, thereby allowing the busbar **30** and the lead tab portion **12** to be connected to each other. In addition, a weld bead **50** in FIG. **6** may be formed between the busbar **30** and the lead tab portion **12** after the welding is finished.

[0065] The guiding portion **300** may contact the welding wire **210** discharged from the supply portion **200** to move the welding wire in the direction in which the lead tab portion **12** extends. The guiding portion **300** may contact the welding wire **210**. One end of the welding wire **210** discharged from the supply portion **200** may contact the guiding portion **300**. To this end, the guiding portion **300** may include a fixing portion **310** formed by depressing one surface. The fixing portion **310** may be formed to face the direction in which the welding wire **210** is supplied.

[0066] One end of the welding wire **210** may be positioned on the fixing portion **310**. The fixing portion **310** may pressurize the welding wire **210** to secure the welding wire **210**. The fixing portion **310** may be removable from the welding wire **210**.

[0067] That is, when the welding wire **210** is discharged from the supply portion **200**, the fixing portion **310** may pressurize and hold the welding wire **210**. With the welding wire **210** secured to the fixing portion **310**, the guiding portion **300** may be moved. When the welding is completed, the welding wire **210** may be disconnected from the fixing portion **310**.

[0068] The fixing portion **310** may be movable in the direction in which the lead tab portion **12** extends. The movement of the fixing portion **310** allows the welding wire **210** to cover the lead tab portion **12**.

[0069] The distance at which the fixing portion **310** is moved may vary depending on the length of the lead tab portion **12** or the area to be welded. More specifically, the fixing portion **310** may move so that the welding wire **210** may be longer than the lead tab portion **12**.

[0070] Referring to FIGS. **4** and **5**, with the lead tab portion **12** secured to the fixing portion **310**, the fixing portion **310** may move downwardly. The welding wire **210** may be positioned to overlap with at least one of the busbar **30** and the lead tab portion **12**. Alternatively, the welding wire **210** may be positioned between the busbar **30** and the lead tab portion **12**. As the welding wire **210** is positioned between the busbar **30** and the lead tab portion **12**, the welding wire **210** may be further applied to weld the lead tab portion **12** even when the lead tab portion **12** has a short protrusion length.

[0071] After the welding wire **210** is positioned between the busbar **30** and the lead tab portion **12**, the laser irradiation portion **400** may irradiate the welding wire **210** with a laser to weld the welding wire **210**. The laser may be irradiated onto the welding wire **210** and onto the busbar **30** and lead tab portion **12** adjacent to the welding wire **210**.

[0072] In an embodiment, the laser may be irradiated in an extension direction of the lead tab portion **12**. In this manner, welding proceeds sequentially from one end to the other end to increase welding stability and performance.

[0073] Referring to FIG. **6**, after the laser is irradiated, the weld bead **50** may be formed. The weld bead **50** may be formed in the region where the welding wire **210** is located. In other words, the weld bead **50** may extend in the longitudinal direction of the lead tab portion **12**.

[0074] The cutting portion **500** may cut the welding wire **210** moving in the direction in which the lead tab portion **12** extends. The cutting portion **500** may be provided as a knife. The present disclosure is not limited thereto. The cutting portion **500** may be configured to cut the welding wire **210**. The cutting portion **500** may cut one end of the welding wire **210** which is connected to the fixing portion **310**.

[0075] The cutting portion **500** may be located on the guiding portion **300**. The cutting portion **500** may be located on the guiding portion **300** and operate independently of the guiding portion **300**. The cutting portion **500** may move in parallel with the protruding direction of the lead tab portion **12** to cut the welding wire **210**.

[0076] Referring to FIG. **6**, the cutting portion **500** may cut a portion of the welding wire **210** which has been completely welded. More specifically, the cutting portion **500** may cut a region of the welding wire **210** which is located at the upper side in the direction in which the lead tab portion **12** extends. As a result, the weld bead **50** may be cut off from the welding wire **210** supplied by the supply portion **200**.

[0077] The cutting portion **500** may cut the welding wire **210** several times. The welding wire **210** may be cut at different heights in the direction in which the lead tab portion **12** extends. By gently cutting one end of the weld bead **50**, injury to the user or damage to the component may be prevented.

[0078] Referring to FIG. **7**, after the welding wire **210** is cut, the fixing portion **310** may fix the welding wire **210** which is supplied again from the supply portion **200**, and the processes described above may be repeated.

[0079] FIGS. **8** and **9** show the supply portion **200** according to one embodiment of the present disclosure.

[0080] More specifically, FIG. **8** shows the components of the supply portion **200** in more detail. The supply portion **200** may receive and discharge the welding wire **210** to the outside. The supply portion **200** may include a winder **230** and rollers **240**. The welding wire **210** is wound around the outer side of the winder **230**, and the rollers **240** unidirectionally feed the welding wire **210** unwound from the winder **230** through rotation with the welding wire **210** interposed between the rollers **240**.

[0081] The winder **230** and the rollers **240** may be provided in the housing **220**. The welding wire **210** which is unwound from the winder **230** may be stably fed out of the housing **220** by the rollers **240** provided in a discharging portion **221**.

[0082] The winder **230** may be rotatably provided. Referring to FIG. **8**, the winder **230** may rotate clockwise about an axis. However, the winder **230** may also rotate counter-clockwise, or the direction of the axis may vary.

[0083] The welding wire **210** may be wound and stored on the winder **230**. The winder **230** may rotate to eject the welding wire **210**. The welding wire **210** may be positioned between the rollers **240**. The rollers **240** may guide the wire **210** being unwound by rotation into the discharging portion **221**.

[0084] The supply portion **200** may be located above the plurality of battery cells **10**. The discharging portion **221** may extend from the housing **220** toward the direction in which the lead tab portion **12** is located. Since it is difficult to control the welding wire **210** when the length of the welding wire **210** increases, the discharging portion **221** may be elongated to guide the welding wire **210** to a desired location.

[0085] The discharging portion **221** may preferably extend to the location of the lead tab portion **12**. The welding wire **210** fed from the discharging portion **221** may be secured by the fixing portion **310** to move the welding wire **210**.

[0086] The battery manufacturing apparatus of the present disclosure may supply the welding wire **210** to each of the lead tab portions **12** provided on the plurality of battery cells **10**. Thus, plurality of supply portions **200** may be provided, or the supply portion **200** may be moved to supply a wire to each of the lead tab portions **12**.

[0087] The supply portion **200** may move in the direction in which the plurality of battery cells **10** are stacked. Once welding of one lead tab of the plurality of battery cells is completed, the supply portion may be moved to weld another lead tab.

[0088] Referring to FIG. **9**, after welding of the lead tab portion **12** on the left side has been completed, the supply portion **200** may be moved to weld the adjacent lead tab portion **12**. In this manner, the volume of the manufacturing apparatus of the present disclosure is reduce and the supply of welding wire **210** is easily supplied.

[0089] The supply portion **200** may be arranged on a transfer rail (not shown) and transferred, or the supply portion **200** may be mounted on a transfer device (not shown) and moved, or transferred by a robot.

[0090] FIGS. **10** and **11** show the blocking portion **600** according to one embodiment of the present disclosure.

[0091] The manufacturing device of the present disclosure may further include the blocking portion **600**. The blocking portion **600** may be positioned between the laser irradiation portion **400** and the plurality of battery cells **10** to cover at least some of the plurality of battery cells **10**. The blocking portion **600** may protect the plurality of battery cells **10** from the laser. In other words, the blocking portion **600** may prevent laser irradiation to areas other than the welded area during laser welding.

[0092] The blocking portion **600** may include a material which absorbs or reflects a laser beam of a particular wavelength.

[0093] Referring to FIG. **10**, in the direction in which the lead tab portion **12** protrudes, the blocking portion **600** may be arranged to overlap the plurality of battery cells **10**. Further, an opening is formed through a region of the blocking portion **600** corresponding to the lead tab and allows the laser to pass therethrough.

[0094] Also, referring to FIG. **11**, the blocking portion **600** may include through-holes **610** which are formed through regions corresponding to the lead tab portions **12**. The through-hole **610** allows the laser to pass therethrough. In other words, the blocking portion **600** may open an area to be welded to allow laser to therethrough and block other areas from the laser.

[0095] The lead tab portion **12**, the busbar **30**, and the welding wire **210** which are opened by the through-hole **610** may be welded by the laser, but other areas may not be irradiated by the laser. Hereinafter, the battery manufacturing method of the present disclosure will be described in detail with reference to FIG. **12**.

[0096] FIG. **12** is a flowchart illustrating a battery manufacturing method according to one embodiment of the present disclosure.

[0097] Referring to FIG. **12**, the battery manufacturing method of the present disclosure includes inserting the lead tab portion **12** into the busbar **30** at step S**10**, supplying the welding wire **210** toward the lead tab portion **12** from the supply portion **200** supplying the welding wire **210** which is at least partially molten and welds the lead tab portion **12** and the busbar **30** at step S**30**, and irradiating the lead tab portion **12**, the busbar **30**, or the welding wire **210** with a laser at step S**70**.

[0098] First, the lead tab portion **12** may be inserted into the busbar **30**. More specifically, the lead tab portion **12** may be inserted into the insertion hole **31** provided in the busbar **30**. However, the lead tab portion **12** may also be inserted into a slit which is open on one side of the busbar **30**, rather than into the insertion hole **31** in FIG. **2**.

[0099] The plurality of battery cells **10** may be secured on a support member (not shown). The busbar **30** may be moved toward the lead tab portions **12** of the plurality of fixed battery cells **10** and inserted into the insertion holes **31**. The busbar **30** may be moved by a separate device (not shown). For example, the busbar **30** may be held and moved by a robot.

[0100] According to the manufacturing method of the present disclosure, the welding wire **210** may be supplied from the supply portion **200** towards the lead tab portion **12** after the lead tab portion **12** is inserted into the busbar **30**. At step S**30** of supplying the welding wire **210**, the welding wire **210** may be moved in the direction in which the lead tab portion **12** extends. In other words, the welding wire **210** is fed in the direction in which the lead tab portion **12** extends so that welding may proceed stably and quickly.

[0101] After the welding wire **210** is supplied, laser irradiation may be performed at step S**70**. In other words, once the welding wire **210** is supplied and positioned adjacent to the lead tab portion **12** and the busbar **30**, welding may proceed by irradiating the welding wire **210** with a laser to melt the welding wire **210**.

[0102] Further, according to the manufacturing method of the present disclosure, the laser may be irradiated in the direction in which the lead tab portion **12** extends during laser irradiation at step S**70**. Eventually, after the welding wire **210** is moved in the direction in which the lead tab portion **12** extends, the laser may be irradiated onto the welding wire **210**.

[0103] Step S**30** in which the welding wire **210** is fed may be performed simultaneously with step S**70** in which the laser is irradiated. In other words, the laser may be irradiated as the welding wire **210** is fed.

[0104] The manufacturing method of the present disclosure may further include arranging the blocking portion **600** at step S**50**. According to the manufacturing method of the present disclosure, the blocking portion **600** covering at least a portion of the plurality of battery cells **10** may be arranged on one side of the plurality of battery cells **10** in the protruding direction of the lead tab portion **12**.

[0105] Step S**50** of arranging the blocking portion **600** may be performed between step S**10** of inserting the lead tab portion **12** into the busbar **30** and step S**70** of irradiating the laser. Further, step S**50** of arranging the blocking portion **600** may preferably be performed after step S**30** in which the welding wire **210** is fed. When the guiding portion **300** moves the welding wire **210**, the blocking portion **600** may restrict the movement radius of the guiding portion **300**.

[0106] The manufacturing method of the present disclosure may further include cutting the welding wire **210** at step S**90**. The manufacturing method of the present disclosure may further include cutting the welding wire **210** moving in the direction in which the lead tab portion **12** extends at step S**90**. The cutting portion **500** may cut the welding wire **210** connected to the guiding portion **300** after the laser is irradiated onto the welding wire **210**. Further, the cutting portion **500** may cut a region of the welding wire **210** which is connected from the top in the direction in which the lead tab portion **12** extends.

[0107] In other embodiments, step S**90** of cutting the welding wire **210** may be performed before and after step S**70** of irradiating the laser. In other words, after the welding wire **210** is fed in the direction in which the lead tab portion **12** extends, step S**90** of cutting the welding wire **210** secured to the guiding portion **300** may be performed, and the laser may then be irradiated. After the laser is irradiated, step S**90** of cutting the welding wire **210** connected to the supply portion **200** may be performed.

[0108] The manufacturing method of the present disclosure may further include moving the supply portion **200** in a direction where the plurality of battery cells **10** are stacked at step S**110**. After the lead tab portions **12** of any one of the plurality of battery cells **10** is completely welded (S**100**), the supply portion **200** may move to weld another lead tab portion **12**.

[0109] The control unit **100** may determine whether each lead tab portion **12** of the plurality of battery cells **10** has been welded. That is, when the control unit **100** determines that welding each lead tab portion **12** is not completed, the manufacturing method of the present disclosure may include moving the supply portion **200** so as to continue welding. On the other hand, when the control unit **100** determines that each lead tab portion **12** is completely welded, the manufacturing method of the present disclosure may be terminated.

[0110] According to one embodiment of the present disclosure, a battery cell and a busbar may be welded quickly and reliably.

[0111] In addition, welding may be performed reliably regardless of a length of a lead tab portion.

[0112] The present disclosure may be modified and implemented in various forms, and its scope is not limited to the above-described embodiments. The content described above is merely an example of applying the principles of the present disclosure, and other features may be further included without departing from the scope of embodiments according to the present disclosure.

What is claimed is:

1. A battery manufacturing apparatus for manufacturing a battery assembly including a plurality of battery cells, each having a lead tab portion electrically connected to an outside and protruding outwardly, and a busbar electrically connecting one or more battery cells among the plurality of battery cells, the battery manufacturing apparatus comprising:

a welding wire, at least a portion of which is melted to weld the lead tab portion and the busbar;

a supply portion supplying the welding wire to the lead tab portion;

a guiding portion contacting the welding wire discharged from the supply portion and moving the welding wire in a direction in which the lead tab portion extends; and

a laser irradiation portion irradiating the lead tab portion, the busbar, or the welding wire with a laser.

2. The battery manufacturing apparatus of claim **1**, wherein the supply portion includes:

a winder around which the welding wire is wound on an outer side thereof; and

a roller unidirectionally transferring the welding wire unwound from the winder by rotation with the welding wire interposed therebetween.

3. The battery manufacturing apparatus of claim **2**, wherein the winder is rotatably provided.

4. The battery manufacturing apparatus of claim **1**, wherein the guiding portion further includes a fixing portion formed by recessing one surface thereof.

5. The battery manufacturing apparatus of claim **4**, wherein the fixing portion is removable from the welding wire.

6. The battery manufacturing apparatus of claim **4**, wherein the fixing portion is movable in the direction in which the lead tab portion extends.

7. The battery manufacturing apparatus of claim **1**, further comprising a cutting portion cutting the welding wire moved in the direction in which the lead tab portion extends.

8. The battery manufacturing apparatus of claim **7**, wherein the cutting portion is located on the guiding portion.

9. The battery manufacturing apparatus of claim **7**, wherein the cutting portion is a knife.

10. The battery manufacturing apparatus of claim **1**, further comprising a blocking portion positioned between the laser irradiation portion and the plurality of battery cells and covering one or more battery cells among the plurality of battery cells.

11. The battery manufacturing apparatus of claim **10**, wherein the blocking portion is formed by penetrating a region corresponding to the lead tab portion.

12. The battery manufacturing apparatus of claim **1**, wherein the supply portion is movable in a direction in which the plurality of battery cells are stacked.

13. A battery manufacturing method of manufacturing a battery assembly including a plurality of battery cells, each having a lead tab portion electrically connected to an outside and protruding from one side thereof, and a busbar electrically connecting one or more battery cells among the plurality of battery cells, the battery manufacturing method comprising:

inserting the lead tab portion into the busbar;

supplying the welding wire toward the lead tab portion from the supply portion supplying the welding wire welding the lead tap portion and the busbar by melting at least a portion of the welding wire; and

irradiating the lead tab portion, the busbar, or the welding wire with a laser.

14. The battery manufacturing method of claim **13**, wherein in the supplying of the welding wire, the welding wire moves in a direction in which the lead tab portion extends.

15. The battery manufacturing method of claim **13**, further comprising moving the supply portion in a direction in which the plurality of battery cells are stacked.

16. The battery manufacturing method of claim **13**, further comprising arranging a blocking portion covering one or more battery cells among the plurality of battery cells on one side of the plurality of battery cells in a protruding direction of the lead tab portion.

17. The battery manufacturing method of claim **13**, wherein the irradiating the laser includes irradiating the laser in a direction in which the lead tab portion extends.

18. The battery manufacturing method of claim **13**, further comprising cutting the welding wire moved in a direction in which the lead tab portion extends.

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